

6.3 Use proper notation

$$\frac{du}{dt} = \begin{bmatrix} a & c & -b \\ c & 0 & a \\ b & -a & 0 \end{bmatrix} u$$

$$a \cdot 2u_1 + 2u_2 + 2u_3$$

$$\Rightarrow 2u_1(u_2 - u_3) + 2u_2(u_3 - u_1) + 2u_3(u_1 - u_2) = 2u_1u_2 - 2u_1u_3 + 2u_2u_3 - 2u_2u_1 - 2u_3u_2 + 2u_3u_1 = 0$$

$$b. Q = e^{At} = \sum \frac{(At)^n}{n!} = I + At + \frac{(At)^2}{2} + \frac{(At)^3}{6} + \dots$$

$$\Rightarrow Q^T = \sum \frac{(A^T t)^n}{n!} = \sum \frac{(A^T)^n t^n}{n!} = I + A^T t + \frac{(A^T)^2 t^2}{2} + \dots$$

Everything cancels except I

$$Q^T Q = I$$

I think solution is wrong

$$24. A = \begin{bmatrix} 1 & 3 \\ 0 & 0 \end{bmatrix} \quad S = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$\lambda_1 = 1, \lambda_2 = 0 \quad \Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$A - I = \begin{bmatrix} 0 & 3 \\ 0 & -1 \end{bmatrix} \Rightarrow \lambda_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 3 \\ 0 & 0 \end{bmatrix} \Rightarrow \lambda_2 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

$$S e^{\Lambda t} S^{-1} = \sum \frac{(A t)^n}{n!} = I + A t + \frac{(A t)^2}{2} + \dots$$

$$= S \left(I + \Lambda t + \frac{\Lambda^2 t^2}{2} + \dots \right) S^{-1}$$

$$= I + A t + \frac{(A t)^2}{2} + \dots$$

$$= e^{At}$$

$$= \begin{bmatrix} e^t & 3(e^t - 1) \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} e^t & 3(e^t - 1) \\ 0 & 1 \end{bmatrix}$$

-2

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow A = U \Lambda U^{-1} \quad U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad U^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} e^t & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} e^t & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} e^t & 0 \\ 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow (1 - \lambda)(1 - \lambda) = 0$$

$$\lambda^2 - 2\lambda + 1 = 0$$

$$\lambda = \frac{2 \pm \sqrt{4 - 4}}{2} = \frac{2 \pm 0}{2} = 1$$

$$\lambda = 1$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^6 = S \Lambda^6 S^{-1} \quad \text{Note } \lambda^2 = \lambda - 1$$

$$\Rightarrow \lambda^3 = \lambda^2 - \lambda$$

$$\Rightarrow \lambda^6 = \lambda^5 - \lambda^4 = 1$$

$$29. A = \begin{bmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 1 \end{bmatrix} \quad A = \begin{bmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 1 \end{bmatrix}$$

$$\Rightarrow (1 - \lambda)(1 - \lambda) + 2 = 0 \quad (1 - \lambda)(1 - \lambda) + 2 = 0$$

$$\lambda^2 - 2\lambda + 3 = 0 \Rightarrow \lambda = 1 \pm i$$

$$A^4 = \begin{bmatrix} -31 & 24\sqrt{2} \\ 24\sqrt{2} & -31 \end{bmatrix}$$

$$\text{with } A - I = \begin{bmatrix} 0 & \sqrt{2} \\ \sqrt{2} & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 - i & \sqrt{2} \\ -\sqrt{2} & 1 - i \end{bmatrix} \Rightarrow \lambda_1 = \begin{bmatrix} 1 \\ i \end{bmatrix} \Rightarrow \lambda_2 = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

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$$30. \begin{bmatrix} 1 & -\alpha/2 \\ \alpha/2 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & \alpha/2 \\ -\alpha/2 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & -\alpha/2 \\ \alpha/2 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & \alpha/2 \\ -\alpha/2 & 1 \end{bmatrix}$$

$$= \frac{1}{1 + \alpha^2/4} \begin{bmatrix} 1 & -\alpha/2 \\ \alpha/2 & 1 \end{bmatrix} \begin{bmatrix} 1 & \alpha/2 \\ -\alpha/2 & 1 \end{bmatrix}$$

$$= \frac{1}{1 + \alpha^2/4} \begin{bmatrix} 1 - \alpha^2/4 & -\alpha/2 + \alpha/2 \\ \alpha/2 - \alpha/2 & 1 - \alpha^2/4 \end{bmatrix}$$

$$= \frac{1}{1 + \alpha^2/4} \begin{bmatrix} 1 - \alpha^2/4 & 0 \\ 0 & 1 - \alpha^2/4 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \text{Solution also skipped computation for } A^T A$$

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$$23. A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A = I \quad \checkmark \quad O \quad \checkmark \quad P_R \quad \checkmark \quad D \quad \checkmark \quad M \quad \checkmark$$

$$B = I \quad \checkmark \quad O \quad \checkmark \quad P_R \quad \checkmark \quad D \quad \checkmark \quad M \quad \checkmark$$

$$A = LU \quad \checkmark \quad QR \quad \checkmark \quad SAS^{-1} \quad \checkmark \quad GAG^T \quad \checkmark$$

$$B = LU \quad \checkmark \quad QR \quad \checkmark \quad SAS^{-1} \quad \checkmark \quad GAG^T \quad \checkmark$$

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6.4

$$7. a. \begin{bmatrix} 1 & 1 \\ b & 1 \end{bmatrix} \quad b = 2$$

$$\Rightarrow \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$$

$$b. \text{ Because } \det \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} < 0,$$

$$\text{implying one } \lambda, \lambda \dots \text{ det}$$

$$\text{must be negative}$$

$$c. \text{ Since it's a } 2 \times 2,$$

$$\text{a negative e-value would}$$

$$\text{result in positive det.}$$

6.3

$$9. \text{ For a matrix } M,$$

$$\text{each column } m_i \in M \text{ must}$$

$$\text{have } \|m_i\| = 1 \quad \forall m_i \in M$$

$$\text{Therefore if we square } m_i$$

$$(m_i)^2 = (1)^2 = 1. \text{ Therefore}$$

$$\|m_i\|^2 = 1 \text{ and } (m_i)^2 \in M^2$$

$$\text{still holds the property.}$$

10. I + assumed that the

det product $x^T x$ will always

remain "real". A quick counter

example using $\lambda = i$ and $b = (i, i)$

$$\Rightarrow A^T A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ i \end{bmatrix} = \begin{bmatrix} i \\ 1 \end{bmatrix}$$

$$x^T A^T A x = \begin{bmatrix} 1 & i \end{bmatrix} \begin{bmatrix} i \\ 1 \end{bmatrix} = 0 \text{ and}$$

$$x^T x = \begin{bmatrix} 1 & i \end{bmatrix} \begin{bmatrix} 1 \\ i \end{bmatrix} = 0$$

In fact, we get the indefinite

form. Signifying that the assumption

is that division yields a real

number / not indeterminate

$$12. B = A - I = \begin{bmatrix} -2 & -3 \\ -2 & -3 \end{bmatrix}$$

$$\left(\frac{-2}{-10} - \lambda \right) \left(\frac{-3}{-10} - \lambda \right) - 0.06 = 0$$

$$\lambda^2 + \frac{1}{2}\lambda = 0 \Rightarrow \lambda = 0, \lambda = -\frac{1}{2}$$

$$A - I \text{ has } \lambda = 0 \text{ since } \det(A - I) = 0.$$

$$\text{Therefore one } \lambda \text{ must be zero.}$$

$$\text{The steady state is } \lambda = 0,$$

$$\text{as } e^{x t} = e^{0 t} = 1$$

$$\text{for } \lambda = 1/2, e^{-1/2 t}$$

$$28. A = \begin{bmatrix} 4 & 2 + i \\ 2 - i & 0 \end{bmatrix} \quad \lambda = 5, \lambda = -1$$

$$= (4 - \lambda)(-\lambda) - 2 = 0 \quad A x = \lambda x$$

$$= \lambda^2 - 4\lambda - 2 = 0 \quad A^T = A$$

$$(\lambda - 5)(\lambda + 1) = 0 \Rightarrow \lambda = 5, \lambda = -1$$

$$\text{Similarly: } \lambda = 5, \lambda = -1$$

$$\Rightarrow \lambda = 5, \lambda = -1$$

$$30. q_{11} = q_{11}^T, q_{11}^T. \text{ We know that}$$

$$q_{11} q_{11}^T \text{ must be the square}$$

$$\text{of an entry from an orthonormal}$$

$$\text{column of which always}$$

$$\|q_{11}\|^2 = 1. \text{ So } q_{11}$$

$$\text{is a component of that}$$

$$\Rightarrow \|q_{11}\| \leq \lambda_{\max}$$

$$16. [SEE cdeE]$$

$$E\text{-values: } \lambda = 1, 0, 2, 0$$